

# Gene Pulser® Electroporation

**Cell Type** Bacterial, gram negative

**Molecules Electroporated** DNA: plasmid DNA with inserts; pUC and pUC derivatives; relaxed circle

**Species Used** *E. coli*, NM522

## Before the Pulse

**Cell Growth Medium** conventional growth medium (NaCl Tryptone); LB broth; yeast extract

**Growth Phase at Harvest** O.D. (600) =Log phase

**Pre-pulse Incubation** 1 min. on ice

**Wash Solution** Distilled water

## The Pulse

**Instruments Used** Gene Pulser® apparatus

**Electroporation Temperature** 0 °C

**Electroporation Medium** 10% glycerol (distilled water)

**Cuvette Gap** 0.1 cm

**Cell Density** 1/500 initial volume

**Voltage** 2.0 kV

**Volume of Cells** 50 µl

**Field Strength** 20 kV/cm

**DNA Concentration** 1 to 10 ng / µl

**DNA Resuspension Buffer** Distilled water

**Capacitor** 25 µF

**Volume of DNA** 10 µl

**Resistor** 200 Ω (Pulse Controller)

## After the Pulse

**Time Constant** 4.2 msec

**Outgrowth Medium** LB growth medium

## Relevant Publications and/or Comments

**Note:** exponential values designated in parentheses.

**LB:** 1% Bacto tryptone, 0.5% Bacto yeast extract, 0.5% NaCl.

**Outgrowth Temperature** 37 °C

**Length of Incubation** 30 min.

**Selection Method or Assay Used** Ampicillin, 40 µg / µl

**Electroporation Efficiency** > 10 (8) transformants / µg DNA

**Per Cent Survival** Not given

**Name of Submitter** Dr. Scott Bowden

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**Survey Number**

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